

[MS-WSTEP]:

WS-Trust X.509v3 Token Enrollment Extensions

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1 Introduction

The WS-Trust X.509v3 Token Enrollment Extensions are extensions of WS-Trust that are used by a system to request that a **certificate** be issued.

The communication is initiated by a requesting client who requests a new certificate, retrieval of an issued certificate, or retrieval of a server certificate. The server processes the request and generates a response based on the request type.

Sections 1.5, 1.8, 1.9, 2, and 3 of this specification are normative. All other sections and examples in this specification are informative.

1.1 Glossary

This document uses the following terms:

Abstract Syntax Notation One (ASN.1): A notation to define complex data types to carry a message, without concern for their binary representation, across a network. ASN.1 defines an encoding to specify the data types with a notation that does not necessarily determine the representation of each value. ASN.1 encoding rules are sets of rules used to transform data that is specified in the ASN.1 language into a standard format that can be decoded on any system that has a decoder based on the same set of rules. ASN.1 and its encoding rules were once part of the same standard. They have since been separated, but it is still common for the terms ASN.1 and Basic Encoding Rules (BER) to be used to mean the same thing, though this is not the case. Different encoding rules can be applied to a given ASN.1 definition. The choice of encoding rules used is an option of the protocol designer. ASN.1 is described in the following specifications: [\[ITUX660\]](#) for general procedures; [\[ITUX680\]](#) for syntax specification; [\[ITUX690\]](#) for the Basic Encoding Rules (BER), Canonical Encoding Rules (CER), and Distinguished Encoding Rules (DER) encoding rules; and [\[ITUX691\]](#) for the Packed Encoding Rules (PER). Further background information on ASN.1 is also available in [DUBUISSON].

certificate: When referring to X.509v3 certificates, that information consists of a public key, a distinguished name (DN) of some entity assumed to have control over the private key corresponding to the public key in the certificate, and some number of other attributes and extensions assumed to relate to the entity thus referenced. Other forms of certificates can bind other pieces of information.

Certificate Management Messages over CMS (CMC): An internet standard for transport mechanisms for CMS [\[RFC2797\]](#).

certification authority (CA): A third party that issues public key certificates. Certificates serve to bind public keys to a user identity. Each user and certification authority (CA) can decide whether to trust another user or CA for a specific purpose, and whether this trust is to be transitive. For more information, see [\[RFC3280\]](#).

Hypertext Transfer Protocol Secure (HTTPS): An extension of HTTP that securely encrypts and decrypts web page requests. In some older protocols, "Hypertext Transfer Protocol over Secure Sockets Layer" is still used (Secure Sockets Layer has been deprecated). For more information, see [\[SSL3\]](#) and [\[RFC5246\]](#).

Public Key Cryptography Standards (PKCS): A group of Public Key Cryptography Standards published by RSA Laboratories.

security token service (STS): A special type of server defined in WS-Trust [\[WSTrust1.3\]](#).

SOAP action: The HTTP request header field used to indicate the intent of the SOAP request, using a URI value. See [\[SOAP1.1\]](#) section 6.1.1 for more information.

SOAP fault: A container for error and status information within a **SOAP message**. See [\[SOAP1.2-1/2007\]](#) section 5.4 for more information.

SOAP message: An **XML** document consisting of a mandatory SOAP envelope, an optional SOAP header, and a mandatory SOAP body. See [\[SOAP1.2-1/2007\]](#) section 5 for more information.

Unicode: A character encoding standard developed by the Unicode Consortium that represents almost all of the written languages of the world. The **Unicode** standard [\[UNICODE5.0.0/2007\]](#) provides three forms (UTF-8, UTF-16, and UTF-32) and seven schemes (UTF-8, UTF-16, UTF-16 BE, UTF-16 LE, UTF-32, UTF-32 LE, and UTF-32 BE).

Web Services Description Language (WSDL): An XML format for describing network services as a set of endpoints that operate on messages that contain either document-oriented or procedure-oriented information. The operations and messages are described abstractly and are bound to a concrete network protocol and message format in order to define an endpoint. Related concrete endpoints are combined into abstract endpoints, which describe a network service. WSDL is extensible, which allows the description of endpoints and their messages regardless of the message formats or network protocols that are used.

X.509: An ITU-T standard for public key infrastructure subsequently adapted by the IETF, as specified in [\[RFC3280\]](#).

XML: The Extensible Markup Language, as described in [\[XML1.0\]](#).

XML namespace: A collection of names that is used to identify elements, types, and attributes in XML documents identified in a URI reference [\[RFC3986\]](#). A combination of XML namespace and local name allows XML documents to use elements, types, and attributes that have the same names but come from different sources. For more information, see [\[XMLNS-2ED\]](#).

XML Schema (XSD): A language that defines the elements, attributes, namespaces, and data types for **XML** documents as defined by [\[XMLSCHEMA1/2\]](#) and [\[XMLSCHEMA2/2\]](#) standards. An XML schema uses **XML** syntax for its language.

MAY, SHOULD, MUST, SHOULD NOT, MUST NOT: These terms (in all caps) are used as defined in [\[RFC2119\]](#). All statements of optional behavior use either MAY, SHOULD, or SHOULD NOT.

1.2 References

Links to a document in the Microsoft Open Specifications library point to the correct section in the most recently published version of the referenced document. However, because individual documents in the library are not updated at the same time, the section numbers in the documents may not match. You can confirm the correct section numbering by checking the [Errata](#).

1.2.1 Normative References

We conduct frequent surveys of the normative references to assure their continued availability. If you have any issue with finding a normative reference, please contact dochelp@microsoft.com. We will assist you in finding the relevant information.

[MS-WCCE] Microsoft Corporation, "[Windows Client Certificate Enrollment Protocol](#)".

[RFC2119] Bradner, S., "Key words for use in RFCs to Indicate Requirement Levels", BCP 14, RFC 2119, March 1997, <https://www.rfc-editor.org/info/rfc2119>

[RFC2797] Myers, M., Liu, X., Schaad, J., and Weinstein, J., "Certificate Management Messages Over CMS", RFC 2797, April 2000, <http://www.rfc-editor.org/info/rfc2797>

[RFC2986] Nystrom, M. and Kaliski, B., "PKCS#10: Certificate Request Syntax Specification", RFC 2986, November 2000, <http://www.rfc-editor.org/info/rfc2986>

[RFC3066] Alvestrand, H., "Tags for the Identification of Languages", BCP 47, RFC 3066, January 2001, <https://www.rfc-editor.org/info/rfc3066>

[RFC3852] Housley, R., "Cryptographic Message Syntax (CMS)", RFC 3852, July 2004, <https://www.rfc-editor.org/info/rfc3852>

[RFC5246] Dierks, T., and Rescorla, E., "The Transport Layer Security (TLS) Protocol Version 1.2", RFC 5246, August 2008, <https://www.rfc-editor.org/info/rfc5246>

[RFC5280] Cooper, D., Santesson, S., Farrell, S., et al., "Internet X.509 Public Key Infrastructure Certificate and Certificate Revocation List (CRL) Profile", RFC 5280, May 2008, <https://www.rfc-editor.org/info/rfc5280>

[WSDL] Christensen, E., Curbera, F., Meredith, G., and Weerawarana, S., "Web Services Description Language (WSDL) 1.1", W3C Note, March 2001, <https://www.w3.org/TR/2001/NOTE-wsdl-20010315>

[WSSUTP] OASIS, "Web Services Security UsernameToken Profile 1.0", OASIS Standard, March 2004, <http://docs.oasis-open.org/wss/2004/01/oasis-200401-wss-username-token-profile-1.0.pdf>

[WSS] OASIS, "Web Services Security: SOAP Message Security 1.1 (WS-Security 2004)", February 2006, <https://www.oasis-open.org/committees/download.php/16790/wss-v1.1-spec-os-SOAPMessageSecurity.pdf>

[WSTrust1.3Schema] OASIS Standard, "WS-Trust 1.3", <http://docs.oasis-open.org/ws-sx/ws-trust/200512/ws-trust-1.3.xsd>

[WSTrust1.3] Lawrence, K., Kaler, C., Nadalin, A., et al., "WS-Trust 1.3", OASIS Standard March 2007, <https://docs.oasis-open.org/ws-sx/ws-trust/200512/ws-trust-1.3-os.html>

[XMLNS] Bray, T., Hollander, D., Layman, A., et al., Eds., "Namespaces in XML 1.0 (Third Edition)", W3C Recommendation, December 2009, <https://www.w3.org/TR/2009/REC-xml-names-20091208/>

[XMLSCHEMA1] Thompson, H., Beech, D., Maloney, M., and Mendelsohn, N., Eds., "XML Schema Part 1: Structures", W3C Recommendation, May 2001, <https://www.w3.org/TR/2001/REC-xmlschema-1-20010502/>

[XMLSCHEMA2] Biron, P.V., Ed. and Malhotra, A., Ed., "XML Schema Part 2: Datatypes", W3C Recommendation, May 2001, <https://www.w3.org/TR/2001/REC-xmlschema-2-20010502/>

1.2.2 Informative References

[DUBUISSON] Dubuisson, O., "ASN.1 Communication between Heterogeneous Systems", Morgan Kaufmann, October 2000, ISBN: 0126333610.

[SCEP] Nourse, A., and Vilhuber, J. Ed., "Cisco Systems' Simple Certificate Enrollment Protocol", April 2009, <http://tools.ietf.org/html/draft-nourse-scep-19>

1.3 Overview

The WS-Trust X.509v3 Token Enrollment Extensions (WSTEP) defines the token enrollment profile for WS-Trust [\[WSTrust1.3\]](#) to allow a client to request **X.509v3 certificates**.

Existing **certificate authorities (CAs)** support **Abstract Syntax Notation One (ASN.1)** formats such as **PKCS#10** ([\[RFC2986\]](#)), **PKCS#7** ([\[RFC3852\]](#)), or **CMC** ([\[RFC2797\]](#)) to encode a certificate request, and those requests are carried in an existing protocol, such as Windows Client Certificate Enrollment Protocol [\[MS-WCCE\]](#) or Cisco's SCEP ([\[SCEP\]](#)). WSTEP also carries those requests from the client to the issuer.

WSTEP provides for issuance, renewal, and delayed-issuance scenarios for X.509v3 digital certificates. The server is known in WS-Trust [WSTrust1.3] terminology as a **Security Token Service (STS)**.

The WS-Trust protocol [WSTrust1.3] definition provides the framework for the STS and for enrollment profile extensions. A typical client interacts with a STS with a request security token (RST) message. The STS responds to a client request security token message with a request security token response (RSTR) or a **SOAP fault**.

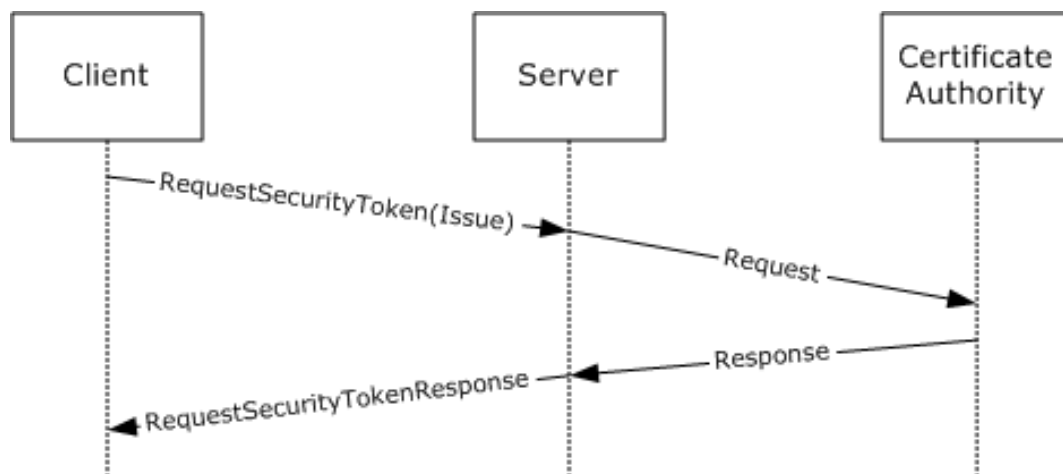


Figure 1: Typical sequence for certificate enrollment

The following figure shows a scenario in which a request cannot be satisfied immediately. In this scenario, the client makes a request, and the server reply indicates that the request is pending some other action. The client then queries the request at a later time, presumably after any conditions for its satisfaction have been met, and receives a reply that the request was issued, rejected, or is still pending.

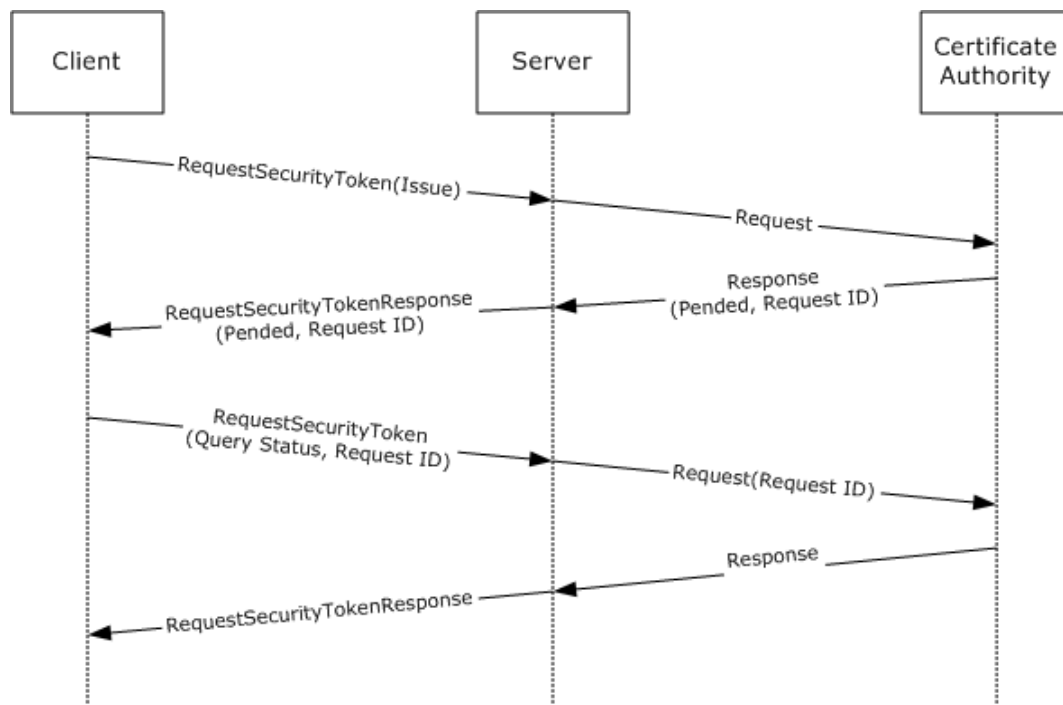


Figure 2: Typical sequence for a pended certificate enrollment request

In some circumstances, the client request could be rejected. In these instances, the STS responds with a SOAP fault. The following figure shows the typical sequence.

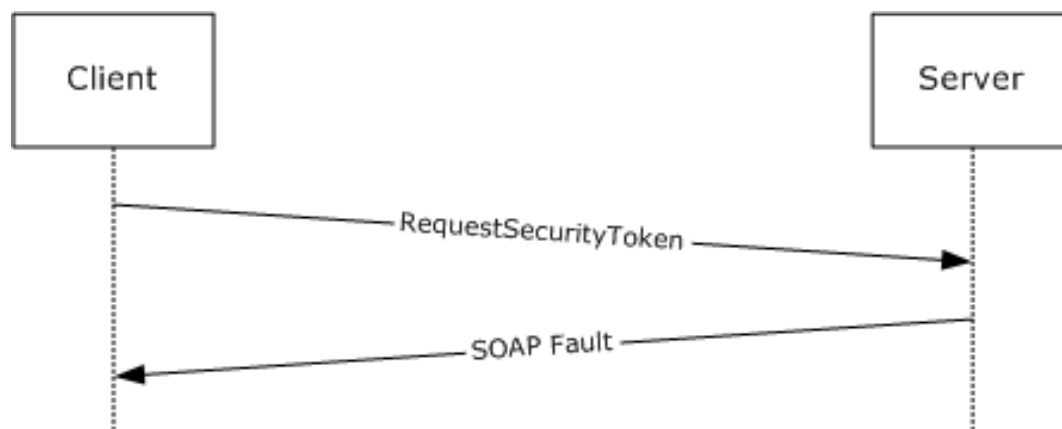


Figure 3: Typical sequence for a rejected certificate renewal request

The following figure is an example of a message exchange for a renewal request. A renewal request uses an existing certificate and requests a new lifespan. From the point of view of the WSTEP protocol, this is the same as an issue request, as the message format is unchanged.

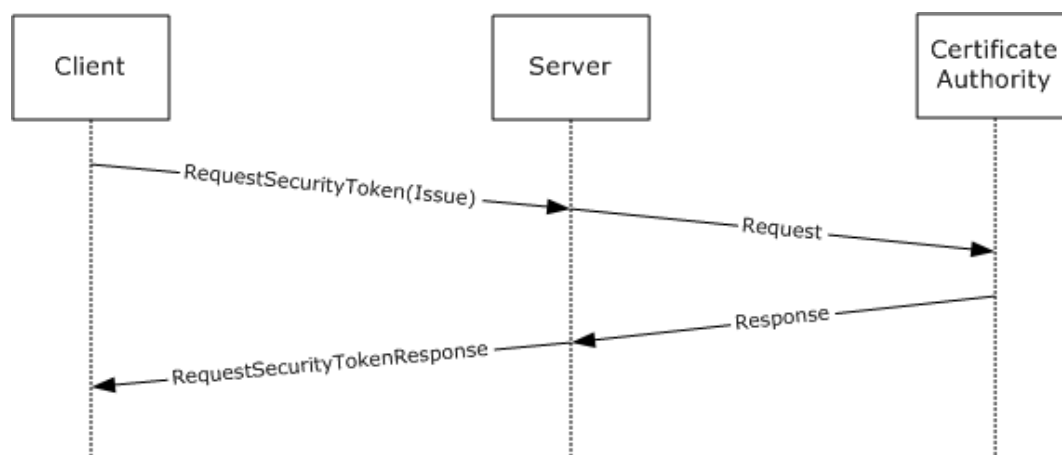


Figure 4: Typical sequence for a certificate renewal request

1.4 Relationship to Other Protocols

The following figure shows the WSTEP Protocol stack diagram.

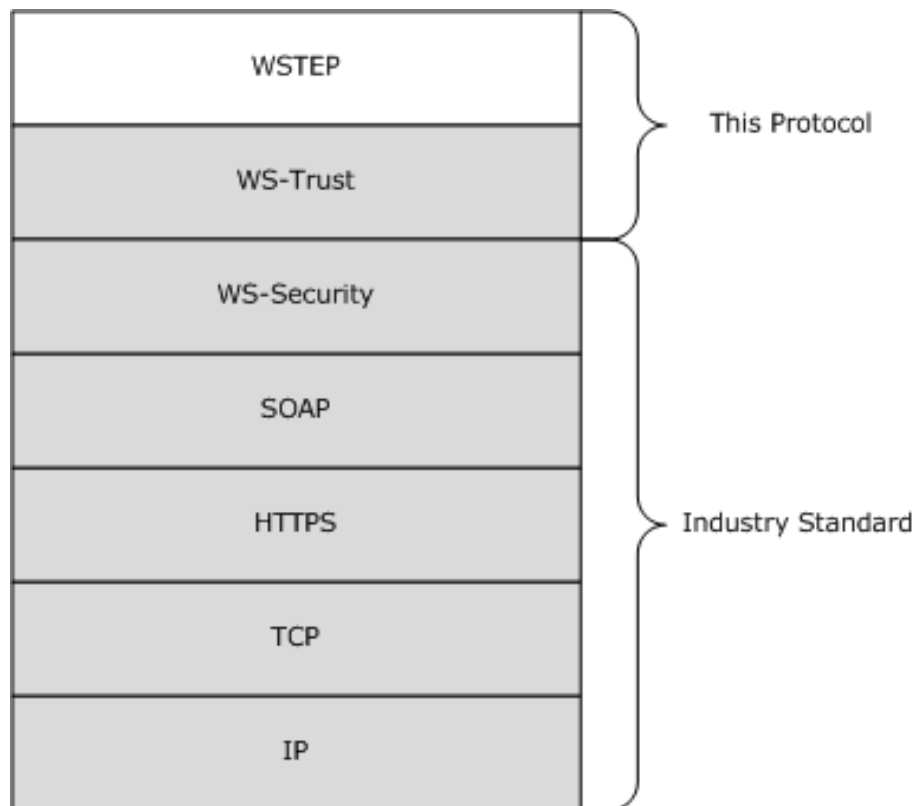


Figure 5: WSTEP Protocol stack diagram

The WSTEP protocol specification is a profile of the WS-Trust Protocol [\[WSTrust1.3\]](#) and makes use of the SOAP and **Hypertext Transfer Protocol over Secure Sockets Layer (HTTPS)** protocols for messaging and security.

1.5 Prerequisites/Preconditions

The WSTEP protocol specification facilitates the issuance of **X.509v3 certificates**. A server implementation of the protocol requires the functionality of a **certificate authority**, capable of interpreting requests in at least one of **PKCS#7**, **PKCS#10**, or **Certificate Management Messages over CMS (CMC)**.

1.6 Applicability Statement

The WSTEP protocol specification is applicable only for requests for **X.509v3 certificates**.

1.7 Versioning and Capability Negotiation

The WSTEP protocol specification does not include versioning and capability negotiation.

1.8 Vendor-Extensible Fields

The WSTEP protocol specification does not include any vendor-extensible fields. WSTEP adheres to the WS-Trust 1.3 [\[WSTrust1.3\]](#) provided extension points.

1.9 Standards Assignments

None.

2 Messages

2.1 Transport

SOAP version 1.2 MUST be used for messaging for the WSTEP protocol. **HTTPS** protocol MUST be used as the transport.

2.2 Common Message Syntax

This section contains common definitions used by this protocol. The syntax of the definitions uses the **XML** schema as defined in [\[XMLSCHEMA1\]](#) and [\[XMLSCHEMA2\]](#), and the **Web Services Description Language (WSDL)** as defined in [\[WSDL\]](#).

2.2.1 Namespaces

This specification defines and references various **XML namespaces**, using the mechanisms specified in [\[XMLNS\]](#). Although this specification associates a specific XML namespace prefix for each XML namespace that is used, the choice of any particular XML namespace prefix is implementation-specific and not significant for interoperability.

Prefixes and XML namespaces used in this specification are as follows.

Prefix	Namespace URI	Reference
xs	http://www.w3.org/2001/XMLSchema	[XMLSCHEMA1]
wst	http://docs.oasis-open.org/ws-sx/ws-trust/200512	[WSTrust1.3]
auth	http://schemas.xmlsoap.org/ws/2006/12/authorization	[XMLSCHEMA1]
wsu	http://docs.oasis-open.org/wss/2004/01/oasis-200401-wss-wssecurity-utility-1.0.xsd	
wsse	http://docs.oasis-open.org/wss/2004/01/oasis-200401-wss-wssecurity-secext-1.0.xsd	
wstep	http://schemas.microsoft.com/windows/pki/2009/01/enrollment	This document

2.2.2 Messages

None.

2.2.3 Elements

This specification does not define any common XML schema element definitions.

2.2.4 Complex Types

This specification does not define any common XML schema complex type definitions.

2.2.5 Simple Types

The WSTEP protocol specification does not define any common XML schema simple type definitions.

2.2.6 Attributes

The WSTEP protocol specification does not define any common XML schema attribute definitions.

2.2.7 Groups

The WSTEP protocol specification does not define any common XML schema group definitions.

2.2.8 Attribute Groups

The WSTEP protocol specification does not define any common XML schema attribute group definitions.

3 Protocol Details

The client side of this protocol is a simple pass-through. No additional timers or other state is required on the client side of this protocol. Calls made by the higher-layer protocol or application are passed directly to the transport layer, and the results returned by the transport are passed directly back to the higher-layer protocol or application.

This section addresses the message processing model for the protocol. It includes related information required by an implementation to successfully send and consume protocol messages.

3.1 SecurityTokenService Server Details

The **SecurityTokenService** hosts a message endpoint that receives **RequestSecurityToken** messages. When received, the server processes the client request and sends it to the **certificate authority**. Upon receiving a response from the certificate authority, a response is generated, and the server sends either a **RequestSecurityTokenResponse** message or a **SOAP fault**. When the message has been sent to the client, the server returns to the waiting state.



Figure 6: Security token service state model

The items of information that are communicated between the server and the certificate authority are specified in this section, but the method of communication used, including timeout and error handling (local API, local remote procedure call (RPC), or some other protocol) is not specified.

The certificate authority MAY have additional requirements that MUST be met in order to issue an **X.509v3 certificate**, such as manager approval, payment processing, or validation of request information. In these instances, a certificate authority response indicating the issuance is pending.

3.1.1 Abstract Data Model

A server supporting the WSTEP protocol maintains a relationship to an issuer which processes messages submitted by the server. When communicating with requestors, a server can support a variety of languages.

Issuer: An address of a **certificate authority (CA)**. The format of the stored address is specific to the implementation and to the form of communication used between the Issuer and the Server.

SupportedLanguages: A list of language identifiers supported by the server. The set of languages are of type xml:lang and defined in [\[RFC3066\]](#).

DefaultLanguage: The default language for the server. DefaultLanguage is of type xml:lang, and the set of supported languages is defined in [\[RFC3066\]](#).

3.1.1.1 Authentication

The WS-Trust X.509v3 Token Enrollment Extensions use the authentication provisions in WS-Security ([\[WSS\]](#)) for the **X.509v3** Security Token issuer to authenticate the X.509v3 Security Token requestor. This section defines the schema used to express the credential descriptor for each supported credential type.

3.1.1.1.1 Kerberos Authentication

Authentication using Kerberos is done at the transport layer.

3.1.1.1.2 X.509v3 Certificate Authentication

Authentication using **X.509v3 certificates** is done at the transport level using Transport Level Security (TLS) 1.2 as defined in [\[RFC5246\]](#).

3.1.1.1.3 Username and Password Authentication

The username and password credential is provided in a request message using the WS-Security Username Token Profile 1.0. The username is provided as defined in section 3.1 of the Ws-Security document [\[WSSUTP\]](#).

3.1.1.1.4 No (Anonymous) Authentication

If no authentication is provided at either the transport layer or the message layer, the request is considered to be anonymous. Anonymous authentication is supported only for renewal requests, where the signature from the existing **certificate** on the request object serves as authentication for the **X.509v3** Security Token requestor.

3.1.2 Timers

None.

3.1.3 Initialization

The *SupportedLanguages* object MUST be initialized with the set of languages that the server supports.

The *DefaultLanguage* parameter MUST be initialized with the language that is to be used by the server when a request does not define a language preference, or the preference is not in *SupportedLanguages*.<1>

3.1.4 Message Processing Events and Sequencing Rules

Operation	Description
wst:RequestSecurityToken2	The wst:RequestSecurityToken2 operation is the sole operation in the WSTEP protocol. It provides the mechanism for certificate enrollment requests, retrieval of pending certificate status, and the request of the server key exchange certificate. The wst:RequestSecurityToken2 operation is defined in WS-Trust 1.3 [WSTrust1.3] .

3.1.4.1 wst:RequestSecurityToken2

The wst:RequestSecurityToken2 operation provides the mechanism for **certificate** enrollment requests, retrieval of pending certificate status, and the request of the server key exchange certificate. The wst:SecurityTokenService port and wst:RequestSecurityToken2 operation are defined in the [\[WSTrust1.3\]](#) **WSDL** wsdl:portType definition.

```
<wsdl:operation name="RequestSecurityToken2">
  <wsdl:input message="wst:RequestSecurityTokenMsg" />
  <wsdl:output message="wst:RequestSecurityTokenResponseCollectionMsg" />
</wsdl:operation>
```

WSTEP makes use of the wst:RequestSecurityToken2 operation. The wst:RequestSecurityToken operation defined in the SecurityTokenService operation is not used. The **wst:RequestSecurityTokenMsg** message consists of a single object definition: the client request. The client request is made using the acceptable **SOAP actions** as defined in section [3.1.4.2](#) and RequestType values, as defined in section [3.1.4.1.2.7](#).

3.1.4.1.1 Messages

The following **WSDL** message definitions are specific to this operation.

3.1.4.1.1.1 wst:RequestSecurityTokenMsg

The wst:RequestSecurityTokenMsg is an incoming message, and is defined in WS-Trust 1.3 [\[WSTrust1.3\]](#) **WSDL**.

wst:RequestSecurityToken: An instance of a **wst:RequestSecurityTokenType** complex type as defined in section [3.1.4.1.3.3](#). The **wst:RequestSecurityToken** element defines the client request and the required information for it to be processed.

3.1.4.1.1.2 wst:RequestSecurityTokenResponseCollectionMsg

The wst:RequestSecurityTokenResponseCollectionMsg is an outgoing message, and is defined in WS-Trust 1.3 [\[WSTrust1.3\]](#) **WSDL**.

wst:RequestSecurityTokenResponseCollectionMsg: An instance of a **wst:RequestSecurityTokenResponseCollection** element as defined in section [3.1.4.1.2.6](#). This element contains the results of the client request.

3.1.4.1.2 Elements

3.1.4.1.2.1 wstep:CertificateEnrollmentWSDetail

The **wstep:CertificateEnrollmentWSDetail** element is used to convey additional information to a client as part of the **SOAP fault** structure when a server returns a SOAP fault.

```
<xs:element name="CertificateEnrollmentWSDetail" nillable="true"
type="wstep:CertificateEnrollmentWSDetailType" />
```

wstep:CertificateEnrollmentWSDetail: An instance of a `<wstep:CertificateEnrollmentWSDetailType>` as defined in section [3.1.4.1.3.7](#). If there is no additional information, the **wstep:CertificateEnrollmentWSDetail** SHOULD be omitted in the SOAP fault.

3.1.4.1.2.2 DispositionMessage

```
<xs:element name="DispositionMessage"
type="wstep:DispositionMessageType" nillable="true" />
```

DispositionMessage: An instance of a `DispositionMessageType` object as defined in section [3.1.4.1.3.1](#).

3.1.4.1.2.3 wst:KeyExchangeToken

The `<wst:KeyExchangeToken>` element is defined in WS-Trust 1.3 [\[WSTrust1.3\]](#) section 8.4.

wst:KeyExchangeToken: The `wst:KeyExchangeToken` element provides a key exchange token that can be used in **certificate** enrollment requests that include the private key.

3.1.4.1.2.4 RequestID

```
<xs:element name="RequestID"
type="xs:string" nillable="true"/>
```

RequestID: A string identifier used to identify a request.

3.1.4.1.2.5 wst:RequestSecurityToken

The `<wst:RequestSecurityToken>` element is defined in WS-Trust 1.3 [\[WSTrust1.3\]](#), section 3.1.

wst:RequestSecurityToken: An instance of a **wst:RequestSecurityTokenType** object as specified in section [3.1.4.1.3.3](#).

3.1.4.1.2.6 RequestSecurityTokenResponseCollection

The `RequestSecurityTokenResponseCollection` is defined in WS-Trust 1.3 [\[WSTrust1.3\]](#), section 3.2.

RequestSecurityTokenResponseCollection: An instance of a **wst:RequestSecurityTokenResponseCollectionType** object as specified in section [3.1.4.1.3.5](#).

3.1.4.1.2.7 wst:RequestType

The `<wst:RequestType>` element is defined in [\[WSTrust1.3\]](#) section 3.1.

wst:RequestType: An instance of a `<wst:RequestTypeOpenEnum>` object as defined in [\[WSTrust1.3\]](#) XML schema definition (XSD).

The <wst:RequestType> MUST have one of the following values:

```
"http://docs.oasis-open.org/ws-sx/ws-trust/200512/Issue"  
"http://schemas.microsoft.com/windows/pki/2009/01/enrollment/QueryTokenStatus"  
"http://docs.oasis-open.org/ws-sx/ws-trust/200512/KET"
```

If the <wst:RequestType> has any other value, the server MUST respond with a **SOAP fault**.

3.1.4.1.2.8 wst:TokenType

The <TokenType> element is defined in [\[WSTrust1.3\]](#), section 3.1.

wst:TokenType: For the **X.509v3** enrollment extension to WS-Trust, the <wst:tokentype> element MUST be <http://docs.oasis-open.org/wss/2004/01/oasis-200401-wss-x509-token-profile-1.0#X509v3>.

3.1.4.1.3 Complex Types

The following XML schema complex type definitions are specific to this operation.

3.1.4.1.3.1 DispositionMessageType

The DispositionMessageType is an extension to the string type that allows an attribute definition of the language for the string. The DispositionMessageType is used to provide additional information about the server processing.

```
<xs:complexType name="DispositionMessageType">  
  <xs:simpleContent>  
    <xs:extension base="xs:string">  
      <xs:attribute ref="xml:lang" use="optional" />  
    </xs:extension>  
  </xs:simpleContent>  
</xs:complexType>
```

xs:string: The string element contains the literal string disposition message returned from the server. The string element contains an xml:lang attribute that defines the language for the string. The language SHOULD be provided for each string element instance.

xml:lang: The language reference xml:lang, indicating the natural or formal language the string element content is written in.

3.1.4.1.3.2 wst:RequestedSecurityTokenType

The wst:RequestedSecurityTokenType is defined in WS-Trust XML schema definition (XSD) [\[WSTrust1.3Schema\]](#).

```
<xs:complexType name="RequestedSecurityTokenType">  
  <xs:sequence>  
    <xs:any namespace="##any" processContents="lax" />  
  </xs:sequence>  
</xs:complexType>
```

MS-WSTEP extends the wst: RequestedSecurityTokenType with two additional elements as follows.

```
<xs:element ref="wsse:BinarySecurityToken" />
```

```
<xs:element ref="wsse:SecurityTokenReference" />
```

The server SHOULD [<2>](#) include the end entity certificate in the **RequestedSecurityToken** response. The ValueType of the **BinarySecurityToken** element for this **RequestedSecurityToken** response MUST be X509v3 [\[RFC5280\]](#). The server MUST also include a CMC full PKI response in the **RequestSecurityTokenResponseCollection**, as specified in sections 4.2 and 4.3 of [\[WSTrust1.3\]](#).

wsse:BinarySecurityToken: The wsse:BinarySecurityToken element contains the issued certificate (2) in either a full CMC response or as a stand-alone x509v3 certificate [\[RFC5280\]](#).

wsse:SecurityTokenReference: A URI reference used to indicate where a pending Certificate Request can be retrieved. The server MUST provide its own URI as the value of the <wsse:BinarySecurityTokenReference:Reference> element as specified in [\[WSTrust1.3\]](#) section 4.2.

3.1.4.1.3.3 wst:RequestSecurityTokenType

The **wst:RequestSecurityTokenType** complex type contains the elements for the security token request in the RequestSecurityTokenMsg message. It is the client-provided object for a **certificate** enrollment request. **wst:RequestSecurityTokenType** is defined in the WS-Trust [\[WSTrust1.3\]](#) XML schema definition (XSD).

```
<xs:complexType name="RequestSecurityTokenType">
  <xs:annotation>
    <xs:documentation>
      Actual content model is non-deterministic, hence wildcard. The following shows intended
      content model:
      <xs:element ref='wst:TokenType' minOccurs='0' />
      <xs:element ref='wst:RequestType' />
      <xs:element ref='wsp:AppliesTo' minOccurs='0' />
      <xs:element ref='wst:Claims' minOccurs='0' />
      <xs:element ref='wst:Entropy' minOccurs='0' />
      <xs:element ref='wst:Lifetime' minOccurs='0' />
      <xs:element ref='wst:AllowPostdating' minOccurs='0' />
      <xs:element ref='wst:Renewing' minOccurs='0' />
      <xs:element ref='wst:OnBehalfOf' minOccurs='0' />
      <xs:element ref='wst:Issuer' minOccurs='0' />
      <xs:element ref='wst:AuthenticationType' minOccurs='0' />
      <xs:element ref='wst:KeyType' minOccurs='0' />
      <xs:element ref='wst:KeySize' minOccurs='0' />
      <xs:element ref='wst:SignatureAlgorithm' minOccurs='0' />
      <xs:element ref='wst:Encryption' minOccurs='0' />
      <xs:element ref='wst:EncryptionAlgorithm' minOccurs='0' />
      <xs:element ref='wst:CanonicalizationAlgorithm' minOccurs='0' />
      <xs:element ref='wst:ProofEncryption' minOccurs='0' />
      <xs:element ref='wst:UseKey' minOccurs='0' />
      <xs:element ref='wst:SignWith' minOccurs='0' />
      <xs:element ref='wst:EncryptWith' minOccurs='0' />
      <xs:element ref='wst:DelegateTo' minOccurs='0' />
      <xs:element ref='wst:Forwardable' minOccurs='0' />
      <xs:element ref='wst:Delegatable' minOccurs='0' />
      <xs:element ref='wsp:Policy' minOccurs='0' />
      <xs:element ref='wsp:PolicyReference' minOccurs='0' />
      <xs:any namespace='##other' processContents='lax' minOccurs='0' maxOccurs='unbounded'
    />
  </xs:documentation>
</xs:annotation>
<xs:sequence>
  <xs:any namespace="##any" processContents="lax" minOccurs="0" maxOccurs="unbounded" />
</xs:sequence>
  <xs:attribute name="Context" type="xs:anyURI" use="optional" />
  <xs:anyAttribute namespace="##other" processContents="lax" />
</xs:complexType>
```

WSTEP extends <wst:RequestSecurityTokenType> with the following elements:

```
<xs:element ref="wsse:BinarySecurityToken" minOccurs="0"
maxOccurs="1" />
<xs:element ref="auth:AdditionalContext" minOccurs="0"
maxOccurs="1" />
<xs:element ref="wst:RequestKET" minOccurs="0" maxOccurs="1" />
<xs:element ref="wstep:RequestID" minOccurs="0" maxOccurs="1" />
```

Only the elements specified below are used in WSTEP. Any element received that is not specified below SHOULD be ignored.

wst:TokenType: Refers to the wst:TokenType definition in section [3.1.4.1.2.8](#).

wst:RequestType: Refers to the wst:RequestType definition in section [3.1.4.1.2.7](#). The wst:RequestType is used to identify the type of the security token request.

wst:RequestKET: Used when requesting a key exchange token as defined in [WSTrust1.3] section 8.4.

wsse:BinarySecurityToken: Provides the DER ASN.1 representation of the certificate request. The type of token is defined by the wst:TokenType element. For the **X.509v3** enrollment extension the wst:TokenType MUST be specified as in section 3.1.4.1.2.8. The certificate request follows the formatting from [\[MS-WCCE\]](#) section 2.2.2.6. The EncodingType attribute of the wsse:BinarySecurityToken element MUST be set to base64Binary.

auth:AdditionalContext: The auth:AdditionalContext element is used to provide extra information in a wst:RequestSecurityToken message. It is an optional element, and SHOULD be omitted if there is no extra information to be passed.

wstep:RequestID: An instance of **wstep:RequestID** as specified in section [3.1.4.1.2.4](#).

3.1.4.1.3.4 wst:RequestSecurityTokenResponseType

The wst:RequestSecurityTokenResponseType contains the elements that are part of a server response to a wst:RequestSecurityToken message. wst:RequestSecurityTokenResponseType is defined in the WS-Trust [\[WSTrust1.3\]](#) XML schema definition (XSD).

```
<xs:complexType name="RequestSecurityTokenResponseType">
  <xs:annotation>
    <xs:documentation>
      Actual content model is non-deterministic, hence wildcard. The following shows intended
      content model:
      <xs:element ref='wst:TokenType' minOccurs='0' />
      <xs:element ref='wst:RequestType' />
      <xs:element ref='wst:RequestedSecurityToken' minOccurs='0' />
      <xs:element ref='wsp:AppliesTo' minOccurs='0' />
      <xs:element ref='wst:RequestedAttachedReference' minOccurs='0' />
      <xs:element ref='wst:RequestedUnattachedReference' minOccurs='0' />
      <xs:element ref='wst:RequestedProofToken' minOccurs='0' />
      <xs:element ref='wst:Entropy' minOccurs='0' />
      <xs:element ref='wst:Lifetime' minOccurs='0' />
      <xs:element ref='wst:Status' minOccurs='0' />
      <xs:element ref='wst:AllowPostdating' minOccurs='0' />
      <xs:element ref='wst:Renewing' minOccurs='0' />
      <xs:element ref='wst:OnBehalfOf' minOccurs='0' />
      <xs:element ref='wst:Issuer' minOccurs='0' />
      <xs:element ref='wst:AuthenticationType' minOccurs='0' />
      <xs:element ref='wst:Authenticator' minOccurs='0' />
      <xs:element ref='wst:KeyType' minOccurs='0' />
      <xs:element ref='wst:KeySize' minOccurs='0' />
      <xs:element ref='wst:SignatureAlgorithm' minOccurs='0' />
```

```

<xs:element ref='wst:Encryption' minOccurs='0' />
<xs:element ref='wst:EncryptionAlgorithm' minOccurs='0' />
<xs:element ref='wst:CanonicalizationAlgorithm' minOccurs='0' />
<xs:element ref='wst:ProofEncryption' minOccurs='0' />
<xs:element ref='wst:UseKey' minOccurs='0' />
<xs:element ref='wst:SignWith' minOccurs='0' />
<xs:element ref='wst:EncryptWith' minOccurs='0' />
<xs:element ref='wst:DelegateTo' minOccurs='0' />
<xs:element ref='wst:Forwardable' minOccurs='0' />
<xs:element ref='wst:Delegatable' minOccurs='0' />
<xs:element ref='wsp:Policy' minOccurs='0' />
<xs:element ref='wsp:PolicyReference' minOccurs='0' />
<xs:any namespace='##other' processContents='lax' minOccurs='0' maxOccurs='unbounded'
/>
</xs:documentation>
</xs:annotation>
<xs:sequence>
<xs:any namespace='##any' processContents='lax' minOccurs='0' maxOccurs='unbounded' />
</xs:sequence>
<xs:attribute name="Context" type="xs:anyURI" use="optional" />
<xs:anyAttribute namespace="##other" processContents="lax" />
</xs:complexType>

```

WSTEP extends the wst:RequestSecurityTokenResponseType with the following elements:

```

<xs:element ref="wstep:DispositionMessage" />
<xs:element ref="wsse:BinarySecurityToken" minOccurs="0" maxOccurs="1" />
<xs:element ref="wstep:RequestID" minOccurs="0" maxOccurs="1" />
<xs:element ref="wst:KeyExchangeToken" minOccurs="0" maxOccurs="1" />
/>

```

Only the elements documented as follows are used by WSTEP. Any element received that is not documented as follows SHOULD be ignored.

wst:TokenType: Refers to the TokenType definition in section [3.1.4.1.2.8](#).

wstep:DispositionMessage: Refers to the definition in section [3.1.4.1.2.2](#). The wstep:DispositionMessage element is used to convey any additional server disposition information as part of the response message.

wsse:BinarySecurityToken: Refers to the wsse:BinarySecurityToken definition in section [3.1.4.1.3.2](#).

wst: KeyExchangeToken: Refers to the wst:KeyExchangeToken definition in section [3.1.4.1.2.3](#).

wst:RequestedSecurityToken: An instance of a wst:RequestedSecurityTokenType object as defined in section 3.1.4.1.3.2.

wstep:RequestID: An instance of a **wstep:RequestID** as defined in section [3.1.4.1.2.4](#) that conveys the request identifier of the originating request.

3.1.4.1.3.5 wst:RequestSecurityTokenResponseCollectionType

The <wst:RequestSecurityTokenResponseCollectionType> is defined in the [\[WSTrust1.3\]](#) XML schema definition (XSD) as a collection of one or more <wst:RequestSecurityTokenResponse> elements. The WS-Trust X.509v3 Token Enrollment Extensions further constrain the [\[WSTrust1.3\]](#) definition and the <wst:RequestSecurityTokenResponseCollection> collection MUST contain at most one <wst:RequestSecurityTokenResponse> element.

```

<xs:complexType name="RequestSecurityTokenResponseCollectionType">
<xs:annotation>

```

```

    <xs:documentation>
The <wst:RequestSecurityTokenResponseCollection> element (RSTRC) MUST be used to return a
security token or response to a security token request on the final
response.</xs:documentation>
    </xs:annotation>
    <xs:sequence>
    <xs:element ref="wst:RequestSecurityTokenResponse" minOccurs="1" maxOccurs="unbounded" />
    </xs:sequence>
    <xs:anyAttribute namespace="##other" processContents="lax" />
</xs:complexType>

```

wst:RequestSecurityTokenResponse: An instance of a [wst:RequestSecurityTokenResponseType](#) object. The <wst:RequestSecurityTokenResponseCollectionType> MUST contain only one <RequestSecurityTokenResponse> element.

3.1.4.1.3.6 wst:RequestTypeEnum

The <wst:RequestTypeEnum> is defined in WS-Trust [\[WSTrust1.3\]](#) XML schema definition (XSD). WSTEP defines the following values for <wst:RequestTypeEnum>.

```
"http://schemas.microsoft.com/windows/pki/2009/01/enrollment/QueryTokenStatus"
```

WSTEP makes use of the Key Exchange Token request type defined in [\[WSTrust1.3\]](#) section 10:

```
"http://docs.oasis-open.org/ws-sx/ws-trust/200512/KET"
```

and the issue request type defined in [\[WSTrust1.3\]](#) XML schema definition (XSD):

```
"http://docs.oasis-open.org/ws-sx/ws-trust/200512/Issue"
```

3.1.4.1.3.7 wstep:CertificateEnrollmentWSDetailType

The <wstep:CertificateEnrollmentWSDetailType> contains additional information pertaining to error conditions.

```

<xs:complexType name="CertificateEnrollmentWSDetailType">
  <xs:sequence>
    <xs:element minOccurs="0" maxOccurs="1" name="BinaryResponse" nillable="true"
type="xs:string" />
    <xs:element minOccurs="0" maxOccurs="1" name="ErrorCode" nillable="true" type="xs:int"
/>
    <xs:element minOccurs="0" maxOccurs="1" name="InvalidRequest" nillable="true"
type="xs:boolean" />
    <xs:element minOccurs="0" maxOccurs="1" name="RequestID" type="xs:string"
nillable="true" />
  </xs:sequence>
</xs:complexType>

```

wstep:BinaryResponse: The wstep:BinaryResponse element is used to provide a response if the Issuer generates one. If there is no response to provide, the wstep:BinaryResponse element MUST be nil.

wstep:ErrorCode: An integer value representing a server error. If there is no error to provide, wstep:ErrorCode MUST be specified as nil.

wstep:InvalidRequest: If the request is denied by the Issuer the server MUST return true. For other errors the wstep:InvalidRequest SHOULD be false.

wstep:RequestID: If the Issuer provides a wstep:RequestID to the server, it MUST be provided to a client. If no wstep:RequestID is provided by the Issuer, the wstep:RequestID element must be specified as nil.

3.1.4.1.4 Attributes

There are no attributes that are specific to this operation.

3.1.4.2 Processing Rules

An incoming **SOAP message** MUST be processed to evaluate the **SOAP actions** and authentication information.

If the user is authenticated successfully using the provided authentication information, message processing MUST continue, and the authentication information SHOULD be provided to the Issuer. If the authentication fails, the server MUST respond with a **SOAP fault**.

If the SOAP action is "http://schemas.microsoft.com/windows/pki/2009/01/enrollment/RST/wstep" the server must follow the Request Security Token Processing Rules per section [3.1.4.2.1](#).

If the SOAP action is "http://docs.oasis-open.org/ws-sx/ws-trust/200512/RST/KET" the server must follow the Key Exchange Token Processing Rules per section [3.1.4.2.2](#).

If any other SOAP action is defined, the server SHOULD respond with a SOAP fault.

3.1.4.2.1 WSTEP Action: Request Security Token Processing Rules

A <wst:RequestSecurityTokenMsg> MUST contain a <wst:RequestType> element as defined in section [3.1.4.1.2.7](#). If the <wst:RequestType> element is absent, nil, or undefined, the server MUST respond with a **SOAP fault**.

If a **wstep:PreferredLanguage** attribute is not present in a <RequestSecurityTokenType> object, or the value is not in SupportedLanguages, the server SHOULD use DefaultLanguage.

If the <wst:RequestType> is "http://docs.oasis-open.org/ws-sx/ws-trust/200512/Issue", the server MUST process the request per section [3.1.4.2.1.1](#).

If the <wst:RequestType> is "http://schemas.microsoft.com/windows/pki/2009/01/enrollment/QueryTokenStatus" the server MUST process the request per section [3.1.4.2.1.2](#).

If the <wst:RequestType> is any other value, the server MUST respond with a SOAP fault.

3.1.4.2.1.1 New and Renewal Request Processing

A wst:RequestSecurityToken message with a wst:RequestType value of "http://docs.oasis-open.org/ws-sx/ws-trust/200512/Issue" is used for the purposes of issuing an **X.509v3 certificate** or for renewal of an existing X.509v3 certificate.

For this type of message, a server has additional syntax constraints on the request message.

wsse:BinarySecurityToken: If the wsse:BinarySecurityToken element is absent or undefined, the server MUST respond with a **SOAP fault**.

wstep:RequestID: If the **wstep:RequestID** element is present and defined, the server SHOULD ignore it.

The server MUST provide the **wsse:BinarySecurityToken** to the Issuer and SHOULD provide the **auth:AdditionalContext** (see section [3.1.4.1.3.3](#)) to the Issuer.

If the Issuer responds with an error, the server MUST respond with a SOAP fault. If the Issuer indicates the issuance is pending, the server MUST use the Issuer response to generate a pending **wst:RequestSecurityTokenResponseCollectionMsg** message. If the Issuer responds with an issued certificate, the server MUST respond with a **wst:RequestSecurityTokenResponseCollectionMsg** message providing the issued certificate.

3.1.4.2.1.2 QueryTokenStatus Request Processing

A **wst:RequestSecurityToken** message with a `<wst:RequestType>` of "http://schemas.microsoft.com/windows/pki/2009/01/enrollment/QueryTokenStatus" is used to retrieve an issued **certificate** or check the status of a certificate request that was pending.

For this type of message, the server has additional syntax constraints on the request message.

The **wstep:RequestID** element is a null-terminated **Unicode** string that contains a certificate request identifier (as defined in section [3.1.4.1.2.4](#)). If the `<wstep:RequestID>` element is absent, defined as nil, or contains no value the server MUST return a **SOAP fault**.

The server MUST provide the **wstep:RequestID** to the Issuer.

If the Issuer responds with an error, the server MUST respond with a SOAP fault. If the Issuer indicates the issuance is pending, the server MUST use the Issuer response to generate a pending **wst:RequestSecurityTokenResponseCollectionMsg** message. If the Issuer responds with an issued certificate, the server MUST respond with a **wst:RequestSecurityTokenResponseCollectionMsg** message providing the issued certificate.

3.1.4.2.2 KET Action: Request Security Token Processing Rules

A **wst:RequestSecurityTokenMsg** MUST contain a `<wst:RequestType>` element as defined in section [3.1.4.1.2.7](#). If the `<wst:RequestType>` element is absent, nil, or undefined, the server MUST respond with a **SOAP fault**.

If the `<wst:RequestType>` is "http://docs.oasis-open.org/ws-sx/ws-trust/200512/KET" the server MUST process the request per section [3.1.4.2.2.1](#).

If the `<wst:RequestType>` is any other value, the server MUST respond with a SOAP fault.

3.1.4.2.2.1 Key Exchange Token Request Processing

A RequestSecurityToken message of `wst:RequestType` of "http://docs.oasis-open.org/ws-sx/ws-trust/200512/KET" is used to retrieve the Key Exchange Token.

For this type of message, a server has additional syntax constraints on the **wst:RequestSecurityTokenMsg** message.

If the `<wst:RequestKET>` element is absent, the server MUST return a **SOAP fault**.

The server requests the Key Exchange Token from the issuer. If the issuer responds with an error, the server MUST respond with a SOAP fault. Otherwise, the server uses the Issuer response to generate a **wst:RequestSecurityTokenResponseCollectionMsg** message.

The `<wst:RequestSecurityTokenResponse>` element in the server response follows the [\[WSTrust1.3\]](#) definition in section 8, but for key exchange in the WSTEP protocol, the `<wst:KeyExchangeToken>` element MUST be present, and provides the key exchange token provided from the Issuer.

3.1.5 Timer Events

None.

3.1.6 Other Local Events

None.

4 Protocol Examples

4.1 RequestSecurityToken Request/Response Message Sequence

In the following message sequence, the username/password authentication headers have been included in the message sequences for clarity.

4.1.1 Standard Certificate Request

4.1.1.1 RequestSecurityToken Message (Issue Request)

```
<s:Envelope xmlns:a="http://www.w3.org/2005/08/addressing"
xmlns:s="http://www.w3.org/2003/05/soap-envelope">
  <s:Header>
    <a:Action s:mustUnderstand="1">
      http://schemas.microsoft.com/windows/pki/2009/01/enrollment/RST/wstep</a:Action>
    <a:MessageID>urn:uuid:b5d1a601-5091-4a7d-b34b-5204c18b5919</a:MessageID>
    <a:ReplyTo>
      <a:Address>http://www.w3.org/2005/08/addressing/anonymous</a:Address>
    </a:ReplyTo>
  </s:Header>
  <s:Body xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xmlns:xsd="http://www.w3.org/2001/XMLSchema">
    <RequestSecurityToken xmlns="http://docs.oasis-open.org/ws-sx/ws-trust/200512">
      <TokenType>http://docs.oasis-open.org/wss/2004/01/oasis-200401-wss-x509-token-profile-
1.0#X509v3</TokenType>
      <RequestType>http://docs.oasis-open.org/ws-sx/ws-trust/200512/Issue</RequestType>
      <BinarySecurityToken EncodingType="http://docs.oasis-open.org/wss/2004/01/
oasis-200401-wss-wssecurity-secext-1.0.xsd#base64binary"
ValueType="http://docs.oasis-open.org/wss/2004/01/
oasis-200401-wss-wssecurity-secext-1.0.xsd#PKCS7"
xmlns="http://docs.oasis-open.org/wss/2004/01/oasis-200401-wss-wssecurity-secext-
1.0.xsd">MIEEDCCAVQCAQAwADCCASIWdQYJKoZIhvcNAQEBBQADgEPADCCAQoCggEBANPk
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      <RequestID xsi:nil="true"
xmlns="http://schemas.microsoft.com/windows/pki/2009/01/enrollment" />
    </RequestSecurityToken>
  </s:Body>
</s:Envelope>
```

4.1.1.2 Server RequestSecurityToken Response

```
<s:Envelope xmlns:s="http://www.w3.org/2003/05/soap-envelope"
xmlns:a="http://www.w3.org/2005/08/addressing">
  <s:Header>
    <a:Action s:mustUnderstand="1">
http://schemas.microsoft.com/windows/pki/2009/01/enrollment/RSTRC/wstep</a:Action>
    <ActivityId CorrelationId="a0f231a3-ccf2-4b9c-99a6-bc353a59b5d0"
xmlns="http://schemas.microsoft.com/2004/09/ServiceModel/Diagnostics">
95427c83-902c-48db-9529-f61cc1d8c035</ActivityId>
    <a:RelatesTo>urn:uuid:b5d1a601-5091-4a7d-b34b-5204c18b5919</a:RelatesTo>
  </s:Header>
  <s:Body>
    <RequestSecurityTokenResponseCollection
xmlns="http://docs.oasis-open.org/ws-sx/ws-trust/200512">
    <RequestSecurityTokenResponse>
    <TokenType>http://docs.oasis-open.org/wss/2004/01/
oasis-200401-wss-x509-token-profile-1.0#X509v3</TokenType>
    <DispositionMessage xml:lang="en-US"
xmlns="http://schemas.microsoft.com/windows/pki/2009/01/enrollment">
Issued</DispositionMessage>
    <BinarySecurityToken
ValueType="http://docs.oasis-open.org/wss/2004/01/
oasis-200401-wss-wssecurity-secext-1.0.xsd#PKCS7"
EncodingType="http://docs.oasis-open.org/wss/2004/01/
oasis-200401-wss-wssecurity-secext-1.0.xsd#base64binary"
xmlns="http://docs.oasis-open.org/wss/2004/01/oasis-200401-wss-wssecurity-secext-
1.0.xsd">MIIRLAYJKoZIHvcNAQCoIIRhTCCEYECAQMxCzAJBgUrDgMCGGUAMH0GCSsGAQUF
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EncodingType="http://docs.oasis-open.org/wss/2004/01/
oasis-200401-wss-wssecurity-secext-1.0.xsd#base64binary"
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1.0.xsd">MIIGijCCBxagAwIBAgIKGNv0wAAAAAPTANBgkqhkiG9w0BAQUFADAzMRswGQYD
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I05oOF8Td5Z2wACKunoU/xrvrWoPoS58TkSDGkd/BWg8R6ZzafdjAF26bGB7HiF5
OMzCxn5Sd/AILOzG3F99/wnqijCxnsFPP6EYrVGs8gjDKfduaIb+cyw4X+wFZch2
DMryOkIN+6YDN60SNPgnHCMq+19P5IjX2aCk3EAyx3NWtf/BOu4KCw==</BinarySecurityToken>

```
eTXL+066yjkNzeK3Dajc/2OT5s/h4Mba6RGMwxyOrFDnQ4HN7uFdB/aGs0QlFN3q
k8g50C5vwpP8OI9hoRjv1c0RB61zXWcuarYZD1iCFwIDAQABo4IDFjCCAxlwRAYJ
KoZihvcNAQkPBDCwNTAObGgghkiG9w0DAgICAIAwDgYIKoZIhvcNAwQCAgCAMAcG
BSs0AwIHMaoGCCqGSIb3DQMHMBcGCSSGAQQBgjcUAgQKHggAVQBzAGUAcjCB2gYI
KwYBBQUHAQEgc0wgcowgccGCCsGAQUFBzAChoG6bGRhcDovLy9DTj1lGQ19FbnRT
dWJDQSxDTj1lBSUesQ049UHVibGljJTIwS2V5JTlWU2VydmljZXMsQ049U2Vydmlj
ZXMsQ049Q29uZmlndXJhdGlvbixEQz1kOS0xMzUxQzA0MDZBLERDPW50dGVzdCxE
Qz1taWNyb3NvZnQsREM9Y29tP2NBQ2VydG1maWNhdGU/YmFzZT9vYmplY3RDdGFz
cz1jZXJ0aWZpY2F0aW9uQXV0aG9yaXR5MB0GA1UdDgQWBRRq9uVvKHZBYbq+35qDh
PiNkw8V7ejAObGNVHQ8BAf8EBAMCBaAwawYDVR0RBGQwYqA3BgorBgEEAYI3FAID
oCkMJ2FiYn1AZDktMTM1MUMwNDA2QS5udHRlc3QubWljcm9zb2Z0LmNvbYEnYWJi
eUEBEOs0xMzUxQzA0MDZBLk5UVEVTVc5NSUNST1NPRlQuQ09NMIHRBgNVHR8EgeMw
geAwgd2ggdggdeGgdRrsZGFwOi8vL0NOPUZCX0VudFN1YkNBLENOPTktMTM1MUMw
NDA3QSxDTj1lDRFASQ049UHVibGljJTIwS2V5JTlWU2VydmljZXMsQ049U2Vydmlj
ZXMsQ049Q29uZmlndXJhdGlvbixEQz1kOS0xMzUxQzA0MDZBLERDPW50dGVzdCxE
Qz1taWNyb3NvZnQsREM9Y29tP2N1cnRpZmljYXRlUmV2b2NhZGlvbKxpc3Q/YmFz
ZT9vYmplY3RDdGFzcz1jUkxEaXN0cmliZXN0cmliZXN0cmliZXN0cmliZXN0cmli
t42RgtEFHdAx4j3213qBdeVcyDapBgNVHSUEIjAgBgorBgEEAYI3CgMEBggrBgEF
BQcDBAYIKwYBBQUHAwIwDQYJKoZIhvcNAQEFBQADggEBAMC53cnFMKGmiFPIrbDo
U4XUoiJnpyM3+Bs+HzihMCRnJdEWh23igwaeXvZcdWejMvuH2xp9qOfLwdOcETH2
Gt9RyJ4Dv/gtlvvaTsbDx10n0fKsH4mAqS6LummQuNBttU0pHJaWb7gqnDhA9gj2
h6HFvX9jhWfDKMovsxbTRWNOAeUx9ovbMgqDlorxeBLG2pkyy26k0XTywsdlc/gQ
f58E3pmc9b3e2QcPPQvanmyOkFW/LnUYtpceTi6kGVLScolqb/984aUeni/Q6G7M
SK4XODae637doX6KZx7pWS4cBUZNhlxkWagWmwzdOpAntale19oMJb+jWw0V3Pzn
xTk=</BinarySecurityToken>
</RequestedSecurityToken>
<RequestID
xmlns="http://schemas.microsoft.com/windows/pki/2009/01/enrollment">61</RequestID>
</RequestSecurityTokenResponse>
</RequestSecurityTokenResponseCollection>
</s:Body>
</s:Envelope>
```

4.1.2 Key Exchange Token Request

4.1.2.1 Client Exchange Token Request

```
<s:Envelope xmlns:a="http://www.w3.org/2005/08/addressing"
xmlns:s="http://www.w3.org/2003/05/soap-envelope">
  <s:Header>
    <a:Action s:mustUnderstand="1">
http://docs.oasis-open.org/ws-sx/ws-trust/200512/RST/KET</a:Action>
    <a:MessageID>urn:uuid:c2884a79-b943-45c6-ac02-7256071de309</a:MessageID>
    <a:ReplyTo>
    <a:Address>http://www.w3.org/2005/08/addressing/anonymous</a:Address>
    </a:ReplyTo>
    </s:Header>
    <s:Body xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xmlns:xsd="http://www.w3.org/2001/XMLSchema">
      <RequestSecurityToken xmlns="http://docs.oasis-open.org/ws-sx/ws-trust/200512">
        <TokenType>http://docs.oasis-open.org/wss/2004/01/oasis-200401-wss-x509-token-profile-
1.0#X509v3</TokenType>
        <RequestType>http://docs.oasis-open.org/ws-sx/ws-trust/200512/KET</RequestType>
        <RequestKET />
        <RequestID xsi:nil="true"
xmlns="http://schemas.microsoft.com/windows/pki/2009/01/enrollment" />
        </RequestSecurityToken>
      </s:Body>
    </s:Envelope>
```

4.1.2.2 Server Key Exchange Token Response

```
<s:Envelope xmlns:s="http://www.w3.org/2003/05/soap-envelope"
xmlns:a="http://www.w3.org/2005/08/addressing">
  <s:Header>
    <a:Action s:mustUnderstand="1">
http://docs.oasis-open.org/ws-sx/ws-trust/200512/RSTR/KETFinal</a:Action>
    <ActivityId CorrelationId="45f6782a-fb93-4a48-b0bb-a21496ba1f3c"
xmlns="http://schemas.microsoft.com/2004/09/ServiceModel/Diagnostics">
17f6073c-c108-4268-9ce4-713ed86894b6</ActivityId>
    <a:RelatesTo>urn:uuid:c2884a79-b943-45c6-ac02-7256071de309</a:RelatesTo>
  </s:Header>
  <s:Body>
    <RequestSecurityTokenResponseCollection
xmlns="http://docs.oasis-open.org/ws-sx/ws-trust/200512">
    <RequestSecurityTokenResponse>
    <TokenType>http://docs.oasis-open.org/wss/2004/01/
oasis-200401-wss-x509-token-profile-1.0#X509v3</TokenType>
    <RequestedSecurityToken>
    <KeyExchangeToken>
    <BinarySecurityToken ValueType="http://docs.oasis-open.org/wss/2004/01/
oasis-200401-wss-x509-token-profile-1.0#X509v3"
EncodingType="http://docs.oasis-open.org/wss/2004/01/
oasis-200401-wss-wssecurity-secext-1.0.xsd#base64binary"
xmlns="http://docs.oasis-open.org/wss/2004/01/oasis-200401-wss-wssecurity-secext-
1.0.xsd">MIIFoJCCBIqgAwIBAgIKGnn1JQAAAAAQDANBgkqhkiG9w0BAQUFADAzMRswGQYD
VQQLExJNANWYyb3NvZnZnQGUETjIFRlYW0xZDASBgNVBAMMC0ZCX0VudFN1YkNBMB4X
DTA5MDMwNTE4MjYyMl0xOTA5MDMwMjE4MzYyMl0wODEbMBkGA1UECzMSTWljcm9z
b2Z0IFBLSzBUZWFtMRkwFwYDVQDDBBGQ19FbnRTdWJDQS1YY2hnMIIBIjANBgkq
hkiG9w0BAQEFAAOCAQ8AMIIBCgKCAQEAWaBSajsw24Kk106+39WQT87+hxahSizX
BXOqJClcZOLjqrSkdc4KnUHV+XXohDO6ETCJ5vkXw9OThT6YWDqpno6G0PJ+h9S3
rmyzlEvXaXg4/eTnDygrVji5QgyXUWK5/BSJFDF160yG2lLlueeS7Eux13Rn12m2
IuvL4OEevhM08XVobhAQmYi1YGkYJImeT2Uq1mVJ0hxjAPI4SY56z2rHdsLFt1Pf
tpQirHPJfWsa3ILMoaw5JODCYf7ixL4IyTaJQJ4+vSttcz0Jyezje0m7mNS8k6aw
P0bzJnGMZkgiq50q9TYN0ZfBYGE0cQRLLyPCITIoav6np0lZEKvCsCQIDAQABo4IC
stCCAq0wHQYDVR0OBBYEFDO96yx8TPm5xHhJkxqrsGmokCeJMB8GA1UdIwQYMBAA
FJ+3jZGCOQUd0DHiPfaXeoF15VzIMIHRBgNVHR8EgeMwgeAwgd2ggdqqgdeGgdRs
ZGFwOi8vL0NOPUZCX0VudFN1YkNBLENOPtktMTM1MUMwNDA3QSxDTj1DRFAsQ049
UHVibGljJTIwS2V5JTlWU2Vydm1jZXMzQ049U2Vydm1jZXMzQ049Q29uZmlndXJh
dGlvbixEQz1kOS0xMzUxQzA0MDZBLERDPW50dGVzdCxEQz1taWNyb3NvZnZnQsREM9
Y29tP2N1cnRpZmljYXRlUmV2b2NhZGlvbKxpc3Q/YmFzZT9vYmplY3RDbGFzZz1j
UkxEaXN0cm1ldXRpb25Qb2ludDCB2gYIKwYBBQUHAQEegc0wgcowgccGCCsGAQUF
BzAChoG6bGRhcDovLy9DTj1GQ19FbnRTdWJDQSxDTj1BSUEsQ049UHVibGljJTIw
S2V5JTlWU2Vydm1jZXMzQ049U2Vydm1jZXMzQ049Q29uZmlndXJhdGlvbixEQz1k
OS0xMzUxQzA0MDZBLERDPW50dGVzdCxEQz1taWNyb3NvZnZnQsREM9Y29tP2NBQ2Vy
dG1maWNhdGU/YmFzZT9vYmplY3RDbGFzZz1jZXJ0aWZpY2F0aW9uQXV0aG9yaXR5
MCMGCCsGAQQBgjcUAQWQHhQAQwBBAEUAeABjAGgAYQBuaGcAZTA3BgkrBgEEAYI3
FQcEKjAoBiARBgEEAYI3FQIDpLIy7eQShOGZGYGtn3+D2dR4gUYBGgIBagIBADAU
BgNVHSUEDTALBgkrBgEEAYI3FQUwDgYDVROPAQH/BAQDAgUgMBwGCCsGAQQBgjcV
CgQPMa0wCwYJKwYBBAGCNxUFMA0GCSqGSIb3DQEBBQUAA4IBAQDESvFy3wAl1BjJ
pcWYC736HTLsu+90215XQvfFvqswJayHqY6aRGvkoWf6qQcm8IJFp2fM/K29ov1o
KEdR1U/zC36TEL2jCxtJAW9/bwA5XEm9Ph+TFBH9focXFCS9FisFuuJzdaL357eI
WXBuKYDgZQXJcl+naKjC+74dKft/T7URU0e/8TRX0FLXJG+7tECNEtSE5/oBMMo
yF+HNUmSjyoXmvZohWB3J7/9ULMpI6lc0BrLVIKghMmCuIDkIuv67WQj/6NfG7uR
shWg/QbRwuEQk2ls9D9dtZwrN7XWgBbNAF6FnwZg7X9GqIDQ9erb6sZPYWg5Gbiz
XVTXYIKj</BinarySecurityToken>
    </KeyExchangeToken>
    </RequestedSecurityToken>
    </RequestSecurityTokenResponse>
    </RequestSecurityTokenResponseCollection>
  </s:Body>
</s:Envelope>
```

4.1.3 Retrieval of a previously pended certificate request with Query Token Status

4.1.3.1 Client Request

```
<s:Envelope xmlns:a="http://www.w3.org/2005/08/addressing"
xmlns:s="http://www.w3.org/2003/05/soap-envelope">
  <s:Header>
    <a:Action s:mustUnderstand="1">
      http://schemas.microsoft.com/windows/pki/2009/01/enrollment/RST/wstep</a:Action>
    <a:MessageID>urn:uuid:ce330bb2-0ca2-473b-a29a-19e9264666ff</a:MessageID>
    <a:ReplyTo>
      <a:Address>http://www.w3.org/2005/08/addressing/anonymous</a:Address>
    </a:ReplyTo>
  </s:Header>
  <s:Body xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xmlns:xsd="http://www.w3.org/2001/XMLSchema">
    <RequestSecurityToken xmlns="http://docs.oasis-open.org/ws-sx/ws-trust/200512">
      <TokenType>http://docs.oasis-open.org/wss/2004/01/oasis-200401-wss-x509-token-profile-
1.0#X509v3</TokenType>

<RequestType>http://schemas.microsoft.com/windows/pki/2009/01/enrollment/QueryTokenStatus</Re
questType>
    <RequestID
xmlns="http://schemas.microsoft.com/windows/pki/2009/01/enrollment">65</RequestID>
  </RequestSecurityToken>
</s:Body>
</s:Envelope>
```

4.1.4 Message exchange with a server fault

4.1.4.1 Client Request

See section [4.1.1.1](#) for an example of a client request.

4.1.4.2 Server Fault Response

```
<s:Envelope xmlns:s="http://www.w3.org/2003/05/soap-envelope"
xmlns:a="http://www.w3.org/2005/08/addressing">
  <s:Header>
    <a:Action s:mustUnderstand="1">http://schemas.microsoft.com/net/2005/12/
windowscommunicationfoundation/dispatcher/fault</a:Action>
    <a:RelatesTo>urn:uuid:ce330bb2-0ca2-473b-a29a-19e9264666ff</a:RelatesTo>
    <ActivityId CorrelationId="4f0e4425-4883-41c1-b704-771135d18f84"
xmlns="http://schemas.microsoft.com/2004/09/ServiceModel/Diagnostics">
eda7e63d-0c42-455d-9c4f-47ab85803a50</ActivityId>
  </s:Header>
  <s:Body>
    <s:Fault>
      <s:Code>
        <s:Value>s:Receiver</s:Value>
        <s:Subcode>
          <s:Value xmlns:a="http://schemas.microsoft.com/net/2005/12/windowscommunicationfoundation/
dispatcher">a:InternalServiceFault</s:Value>
        </s:Subcode>
      </s:Code>
      <s:Reason>
        <s:Text xml:lang="en-US">The server was unable to process the request
due to an internal error. For more information about the error, either turn
on IncludeExceptionDetailInFaults (either from ServiceBehaviorAttribute or
from the <serviceDebug> configuration behavior) on the server in order to
```


MDQwNkEuTlRURVNULk1JQ1JPU09GVC5DT00wggEiMA0GCSqGSIB3DQEBAQUAA4IB
DwAwggEKAoIBAQCUIUF1eKRjXgCHj0u0lmiL+Gq1uG85wgfsz2th+w0jM+BA+1KL
e57dbCc+FqzpZqJruPgGDSAfMP4o6Kk8roM/4kPEVSJYIBidnC3hRx2txSR7HrcS
Lo8/xhnXWY7m8WjpcFro2mBV/JbOnTT5KFU0Z+YSSCGzEVahJqN2Wj1l3VBZ8YC
J3BEUWY1UDYp33zDnPAMULKDPUPJlMXUmlX+pUL4vycfnmlon4iGw0kHHCqfM7L
NPYJfKdZSgeTfRd2qSPdfUeOurwoS8whyFvPe2LFT5BRcAoF4dIaRK5DYSCP8yv1
xQ+6z/yqD+tZ9WpROTC7gFilxeHPrU3TpBq5AgMBAAGjggMWMIIDEjBEBgkqhkiG
9w0BCQ8ENzA1MA4GCCqGSIB3DQMCAGIAgDAOBggqhkkiG9w0DBAICAIaWbWYFKw4D
AgcwCgYIKoZIhvcNAwcwFwYJKwYBBAGCNxQCBAoeCABVAHMAZQByMIHaBggrBgEF
BQcBAQSBzTCByjCBxwYIKwYBBQUHMAKGbbsZGFwOi8vL0NOPUZCX0VudFN1YkNB
LENOPUFJQSxDTj1QdWJsaWMLmJBLZXklmJBTZXJ2aWNlcYxDTj1TZXJ2aWNlcYxDT
Tj1Db25maWdlcmF0aW9uLERDPWQ5LTEzNTFDMQwNkEsREM9bnR0ZXN0LERDPWlp
Y3Jvc29mdCxEQz1jb20/Y0FDZXJ0aWZpY2F0ZT9iYXNlP29iamVjdENsYXNzPWNl
cnRpZmljYXRpb25BdXRob3JpdHkwHQYDVR0OBByEFAiBMzPGORkGpoPN1ZHy7InY
IlG+MA4GA1UdDwEB/wQEAWIFoDbrBgNVHREEZDBioDcGCisGAQQBgjcUAgoGKQwn
YWJieUBkOS0xMzUxQzA0MDZBLm50dGVzdC5taWNyb3NvZnQuY29tgSdhYmJ5QE
Q5LTEzNTFDMQwNkEuTlRURVNULk1JQ1JPU09GVC5DT00wgesGA1UdHwSB4zCB4DCB
3aCB2qCB14aB1GxkYXA6Ly8vQ049RkRFRW50U3ViQ0EsQ049OS0xMzUxQzA0MDdB
LENOPUNEUCxDTj1QdWJsaWMLmJBLZXklmJBTZXJ2aWNlcYxDTj1TZXJ2aWNlcYxDT
Tj1Db25maWdlcmF0aW9uLERDPWQ5LTEzNTFDMQwNkEsREM9bnR0ZXN0LERDPWlp
Y3Jvc29mdCxEQz1jb20/Y2VydGlmawNhdGVSSXZvY2F0aW9uTGlzdD9iYXNlP29i
amVjdENsYXNzPWNSTERpc3RyaWY1dGlvb1BvaW50MB8GA1UdIwQYMBaAFJ+3jZGC
0QUd0DHiPfaXeoF15VzIMCKGA1UdJQqiMCAGCisGAQQBgjcKAwQGCCsGAQUFBwME
BggrBgEFBQcDAjANBgkqhkiG9w0BAQUFAAOCAQEAge5W3XJ3766qGf8Y+r2Yu1Gm
Q0cskpa7tqeZXwZW5lskiv6i0S1NPo67HNpO8+UfLyRpdghF+ZwtphnCdtPieAVn
0Hmd1LvO+LxTGA16ovCixa3qRG1x2NUQLH7VzH34ugf/YEaJDPJ/9818TJhQQ7iP
Ti3jlu9e52rYmNLb450egdmVJhbKJIftpcQqbWZY7HP3tzizf50dXhhzg3CGkOJI
WqSQPQQRKnUkvxCjvFIIOVrvNM3IlUGDjIu9Ejj1gE1ZSqa4Lixjko2M/hzhWpej
KnVAEV9eb+ppWh25jbIEvk6WryMgnwPgS9gRBB150q01ahPi6Vj5EOFXoe99KTAN
BgkqhkiG9w0BAQUFAAOCAQEAL89mjFAUL4Bi2e/8KvIXY6r8CLESXVUUsNNv16r4
wPdkWeuGNteclvyufhxr3jNAEVwaechKQcIGKNz5cyd6fBE4LDPP+fiI4Ywb1bPU
gJVozQpJV5j2+dSux5TiJQYYeXlqrQfG3ioz7m7WSOvoFlmBomGpFUmH2MMdsVSj
xFvJ7DSHQeQJOMRKc5G1Xp7o5/W5Xi6bn4T0RTIGd6vtsbVAta1DJ/jais6urlv8
yqedfA62BbbEwIOA6wgdhCLLUXilPv3eZy5Ssy2dDBwbkJsVfEhrOyivq6q
0ErE0fcDm/WBqOwhifUgJwIUF9DtE7vykt7n043VZYQLTaCCBPiWggaOMIIFdqAD
AgECAgoY2d8GAAAAA+MA0GCSqGSIB3DQEBBQUAMDMxGzAZBgNVBASTEklpY3Jv
c29mdCBQ80kgVGvHbTEUMBIGAlUEAwWLRkJFRW50U3ViQ0EwHhcNMdkmZa1M1Tgy
nJ3BWhcNMTAwMzA1M1TgyNjE3WjCBvjETMBEGCgmSjOMT8ixkARkWA2NvbTEZMBcG
CgmSjOMT8ixkARkWCWlpY3Jvc29mdDEWMBQGCgmSjOMT8ixkARkWBm50dGVzdDEd
MBsGCgmSjOMT8ixkARkWDWQ5LTEzNTFDMQwNkEsREM9bnR0ZXN0LERDPWlpY3Jvc29m
dCxEQz1jb20/Y0FDZXJ0aWZpY2F0ZT9iYXNlP29iamVjdENsYXNzPWNlcnRpZmlj
YXRpb25BdXRob3JpdHkwHQYDVR0OBByEFAiBMzPGORkGpoPN1ZHy7InYIlG+MA4G
A1UdDwEB/wQEAWIFoDbrBgNVHREEZDBioDcGCisGAQQBgjcUAgoGKQwnYWJieUBk
OS0xMzUxQzA0MDZBLm50dGVzdC5taWNyb3NvZnQuY29tgSdhYmJ5QEQ5LTEzNTFD
MDQwNkEuTlRURVNULk1JQ1JPU09GVC5DT00wgesGA1UdHwSB4zCB4DCB3aCB2qCB
14aB1GxkYXA6Ly8vQ049RkRFRW50U3ViQ0EsQ049OS0xMzUxQzA0MDdBLENOPUNE
UCxDTj1QdWJsaWMLmJBLZXklmJBTZXJ2aWNlcYxDTj1TZXJ2aWNlcYxDTj1Db25m
aWdlcmF0aW9uLERDPWQ5LTEzNTFDMQwNkEsREM9bnR0ZXN0LERDPWlpY3Jvc29m
dCxEQz1jb20/Y2VydGlmawNhdGVSSXZvY2F0aW9uTGlzdD9iYXNlP29iamVjdENs
YXNzPWNSTERpc3RyaWY1dGlvb1BvaW50MB8GA1UdIwQYMBaAFJ+3jZGC0QUd0DHi
PfaXeoF15VzIMCKGA1UdJQqiMCAGCisGAQQBgjcKAwQGCCsGAQUFBwMEBggrBgEF
BQcDAjANBgkqhkiG9w0BAQUFAAOCAQEAge5W3XJ3766qGf8Y+r2Yu1GmQ0cskpa7
tqeZXwZW5lskiv6i0S1NPo67HNpO8+UfLyRpdghF+ZwtphnCdtPieAVn0Hmd1LvO
+LxTGA16ovCixa3qRG1x2NUQLH7VzH34ugf/YEaJDPJ/9818TJhQQ7iPTi3jlu9e
52rYmNLb450egdmVJhbKJIftpcQqbWZY7HP3tzizf50dXhhzg3CGkOJIWqSQPQQR
KnUkvxCjvFIIOVrvNM3IlUGDjIu9Ejj1gE1ZSqa4Lixjko2M/hzhWpejKnVAEV9e
b+ppWh25jbIEvk6WryMgnwPgS9gRBB150q01ahPi6Vj5EOFXoe99KTGCAWgwgGfK

```

AgEBMEEWmZebMBkGAlUECxmSTWljcm9zb2Z0IFBLSSBUZWFtMRQwEgYDVQDDAtG
Ql9FbnRtdWJDQQIKGNfBgAAAAAPjAJBgUrDgMCGGUAMA0GCSqGSIsb3DQEBAQUA
BIIBAF06/80HTK6v4fX5rYijOEpz43tvLQOk/0SfXeg4Nlm47SAzqDzNSZ3QljLJ
vZoBnz4E2vclITZsLYpMN0o4rxflZwc+2X7MtoYbnbmVllZnTnQINDfbmIiXyi+L
zkjw+ZOTZUxqNIXhevKru3P3nDhFENhhSm/qC5Wovg7igCsDh9XJ/G6ZkQ8SEbl
vkBU2lRjpOyKYaEUXz/Y0yViIxpYCPFrByDU50ngXhwOhBcbAc5RImhI8O7xE04W
YQl3sBxWllsFuxMsmzwQlTJrFauvjoPt96Hflog96p9w8DlZKxtlhqCI+XqIIqur
30aWtKmxTQTxG8uBCrczYAgfWGk=</BinarySecurityToken>
  <RequestID xsi:nil="true"
xmlns="http://schemas.microsoft.com/windows/pki/2009/01/enrollment" />
</RequestSecurityToken>
</s:Body>
</s:Envelope>

```

4.1.5.2 Server Request Security Token Response

```

<s:Envelope xmlns:s="http://www.w3.org/2003/05/soap-envelope"
xmlns:a="http://www.w3.org/2005/08/addressing">
  <s:Header>
    <a:Action s:mustUnderstand="1">
http://schemas.microsoft.com/windows/pki/2009/01/enrollment/RSTRC/wstep</a:Action>
    <ActivityId CorrelationId="b17bfb40-747b-477a-a83c-175624e401aa"
xmlns="http://schemas.microsoft.com/2004/09/ServiceModel/Diagnostics">
0a9f1849-8211-489c-a2b7-6a07edle6832</ActivityId>
    <a:RelatesTo>urn:uuid:b0a9b388-2581-451d-8c03-270d4ffe2928</a:RelatesTo>
  </s:Header>
  <s:Body>
    <RequestSecurityTokenResponseCollection
xmlns="http://docs.oasis-open.org/ws-sx/ws-trust/200512">
    <RequestSecurityTokenResponse>
    <TokenType>
http://docs.oasis-open.org/wss/2004/01/
oasis-200401-wss-x509-token-profile-1.0#X509v3</TokenType>
    <DispositionMessage xml:lang="en-US"
xmlns="http://schemas.microsoft.com/windows/pki/2009/01/
enrollment">Issued</DispositionMessage>
    <BinarySecurityToken
ValueType="http://docs.oasis-open.org/wss/2004/01/
oasis-200401-wss-wssecurity-secext-1.0.xsd#PKCS7"
EncodingType="http://docs.oasis-open.org/wss/2004/01/
oasis-200401-wss-wssecurity-secext-1.0.xsd#base64binary"
xmlns="http://docs.oasis-open.org/wss/2004/01/oasis-200401-wss-wssecurity-secext-
1.0.xsd">MIIRLAYJKoZiHvcNAQCoIIRhTCCEYECAQMxCzAJBgUrDgMCGGUAMH0GCCsGAQUF
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AAIwDQYJKoZiHvcNAQEFBQAwNDEbMBkGAlUECxmSTWljcm9zb2Z0IFBLSSBUZWFt

```

MRUwEwYDVQDDAxGQl9FbnRSb290Q0EwHhcNMDkwMzAzMDMyNjE2WhcNMTEwMzAzMDMzNjE2WjAzMRswGQYDVQLExJNaWNyb3NvZnQgUETJIFRlYW0xWDASBgNVBAMMC0ZCX0VudFNlYkNBMIIBIjANBgkqhkiG9w0BAQEFAAOCAQ8AMIIBCgKCAQEazqt18Vme3t1durXs81ORjWBWoxDQTTPJAlYNQdqvS+H2HutrUNjvW/+vKOAm0ib8GR3uD8IT+Kk8TJvZSGOuQAKxYtzaqjDt7Alc7Utsne1SiKDT5ZSflpmfUvASKd28jJ4YB1SDJSiJmOyWqUzCnwAwW0VXCrMklQnyGj3Akq+p6Mgo/ZqaeFuJ4o7jJjI/emBN5mM0lX00yjqP79bQbJagd0bidQkfsASDW3HAycTcWuKfmfjF5vPMZqLWcaRjASv2tEn9urPDJv4mhXrTz53ETuplKaOmA95BQULnLXXHreXImUkGacdPhdlhCWm9KEv1Qlvsks9UUBfHucKwIDAQAB04ICXzCCAlswEAYJKwYBBAGCNxUBBAMCAQAwHQYDVR0OBYYEFfJ+3jZGC0QUd0DHiPfaxeof15VzIMBkGCSsGAQQBgjcUAgQMHgoAUwB1AGIAQwBBMA4GA1UdDwEB/wQEAWIBhjAPBgNVHRMBAf8EBTADAQH/MB8GA1UdIwQYMBAAFL2g1sm5bP3a8u5BIgmUPNqaDd1VMIHsBgNVHR8EgeQwgeEwgd6ggduggdiGgdVsZGFwOi8vLONOPUZCX0VudFJvb3RDQSxDtj05LTEzNTFDMQWnKESQ049Q0RQLENOPVB1YmXpYyUyMetleSUyMFN1cnZpY2VzLENOPVN1cnZpY2VzLENOPUNvbmZpZ3VyYXRpb24sREM9ZDktMTM1MUMwNDA2QSxQZ1udHRlc3QsREM9bWljcm9zb2Z0LERDPWNvbT9jZXJ0aWZpY2F0ZVJldm9jYXRpb25MaXN0P2Jhc2U/b2JqZWNOQ2Zxc3M9Y1JMRGlzdhJpYnV0aW9uUG9pbmQwgdSGCCsGAQUFBwEBBIIHOMIHLMIHIBggrBgEFBQcwAoBuZ2xkYXA6Ly8v49RkJfFRW50Um9vdENBLENOPUFJQSxDtj1QdWJsawW1MjBLZ2XklMjBTZXJ2aWN1cyxDtj1TZXJ2aWN1cyxDtj1Db25maWdlcmF0aW9uLERDPWQ5LTEzNTFDMQWnKESREM9bnR0ZXN0LERDPWlY3Jvc29mdCxEQZ1jb20/Y0FDZXJ0aWZpY2F0ZT9iYXNlP29iamVjdENsYXNzPWN1cnRpZmljYXRpb25BdXRoY3JpdHkwDjJkOzIhvcNMTAwMzAlMTgyNjIwYjCBvjETMBEGCgmSjOMT8ixkARkWA2NvbTEZMBcGCgmSjOMT8ixkARkWCWlpY3Jvc29mdDEWMBQGCgmSjOMT8ixkARkWbm50dGVzdDEdMBsGCgmSjOMT8ixkARkWdWQ5LTEzNTFDMQWnKESdJAMBgnVBAMTBVZzZXJzMQ0wCwYDVQDEwRhYmJ5MTYwNAJYKozIhvcNAQKBFIhYmJ5QEQ5LTEzNTFDMQWnKESuT1RURVNUk1JQlJPU09GVC5DT00wggEiMA0GCSqGSIb3DQEBAQUAA4IBDwAwggEKAoIBAQQDg7WYEmBjrr1iF0S4zEY2JZG7yTThp8cXI5OLdYS0bwXJLZJWJ3fmTD646r/oCeGKi0oqMJOTJUeMgb0F70fcmJH7GaXe6i+QGPy7DaTYYCtn94wP2QQGK6Mrnr1jPqYUU/1IOVQxukjZnzT11y9E/XfPLoqTm6p3F6GksLe0kt0MIq0xqXV181Hh5mzR7ddrZ4UJyQJ2F00n1qNe233vHmiJbLTR1WFN4a+onBSFAUINtyJXquNdDgza8eyNWeleJ1JzxWbGtGjPhXrjL8wqpxOCs1VgOvdDEDU3mCoCaOLw4i5pURRnj6RN8VemOIVQSB/XR7si3Xfi5wauKNC6rAgMBAAGgggMWMiIDEjBEBgkqhkiG9w0BQ8ENzAlMA4GCCqGSIb3DQMCAGIAgDAOBggqhkiG9w0DBAICAIAwBwYFKw4DAgcwCgYIKoZIhvcNAQwBFjYKwYBBAGCNxQCBaoeCABVAHMAZQBvMIHaBggrBgEFBQcBAQSBzTCBvjCBxwYIKwYBBQUHMAKGbbsZGFwOi8vLONOPUZCX0VudFNlYkNBLENOPUFJQSxDtj1QdWJsawW1MjBLZ2XklMjBTZXJ2aWN1cyxDtj1TZXJ2aWN1cyxDtj1Db25maWdlcmF0aW9uLERDPWQ5LTEzNTFDMQWnKESREM9bnR0ZXN0LERDPWlY3Jvc29mdCxEQZ1jb20/Y0FDZXJ0aWZpY2F0ZT9iYXNlP29iamVjdENsYXNzPWN1cnRpZmljYXRpb25BdXRob3JpdHkwHQYDVROBBYEFfPaFPdU+BIw69/vvrdhlF+E9gRnTMA4GA1UdDwEB/wQEAWIFoDBrBgNVHREEZDBioDcGCisGAQQBgjcUAgOgKQwnYWJieUBkOS0xMzUxQzA0MDZBLm50dGVzdC5taWNyb3NvZnQuY29t9tSdhYmJ5QEQ5LTEzNTFDMQWnKESuT1RURVNUk1JQlJPU09GVC5DT00wgesGA1UdHwSB4zCB4DCB3aCB2qCB14aB1GxkYXA6Ly8v49RkJfFRW50U3ViQ0ESQ049OS0xMzUxQzA0MDdBLENOPUNEUCxDtj1QdWJsawW1MjBLZ2XklMjBTZXJ2aWN1cyxDtj1TZXJ2aWN1cyxDtj1Db25maWdlcmF0aW9uLERDPWQ5LTEzNTFDMQWnKESREM9bnR0ZXN0LERDPWlY3Jvc29mdCxEQZ1jb20/Y2VydG1maWNhdGVScXZvY2F0aW9uTG1zdD9iYXNlP29iamVjdENsYXNzPWNSTERpc3RyaWJldGlvb1BvaW50MB8GA1UdIwQYMBAAfJ+3jZGC0QUd0DHiPfaxeof15VzIMCKGA1UdJQqiMCAGCisGAQQBgjcKAwQGCSsGAQUFBwMEBggrBgEFBQcDAjANBgkqhkiG9w0BAQUFAAOCAQEAoZeiMPDULBva9GIOi/Y4S4EN9x6CMpb26krCUPEL2bRylJ7XJxxmq5VD6Fds9OjdTHkLng4U5TtW6B1cjC55JiaUsQMjEtyr8/b08Xvbg4FbK6MDLHI05hxQSCxNoubX4hjKyG9guJbTau9zLVVziHO7YXzYEAJ95NiQKwG+bIks/uLnF0HjOGq9NE2q6CEDpduP3X4N3X6WjDuZL8aj42GipN9PW9emOn2fwhQz05VT4KIM6y+PJc93B8k3wayStolsGKHChb80Trj6jFLUR7Rtn5LuN7MnnUALB/xsEP/T4iaFSYaPmpLikaRJEwqJAvkQ/g7pmGrYNJ2IOXww/4jGCAakwggG1AgEBMEIwNDEBMBkGA1UECXMSTWljcm9zb2Z0IFBLSSBUZWFtMRUwEwYDVQDDAxGQl9FbnRSb290Q0ECCCEmFNAwAAAAAAIwCQYFKw4DAhoFAKA+MBcGCSqGSIb3DQEJAZEKBggrBgEFBQcMAZajBgkqhkiG9w0BCQQxFgQUoOWV5P4v001TWl71ZHHOacJ3I2QwDQYJKoZIhvcNAQEBBQAEggEAN0UXGUWciZM9zOIFJt50AkL82/DHk43tPELklvP4s3s/7EtKn04qbFVV4//HYWpcXz24dJ/SWZSTApWlKfFKIY6Y67nL/dsnswYSeBWxhzp6ZiYk8klepCR+ebR8Nr1G9ZTow7yE0e9NZW/OR9IifwvPf8EMJBo/6kXZaY/Ppz3ONljXfVmuAipYHn033A8PgR2i2/fjo+z2MC5grmkFOHxd4Sg90inQ

5 Security

5.1 Security Considerations for Implementers

None.

5.2 Index of Security Parameters

None.

6 Appendix A: Full WSDL

The WSTEP protocol is a profile extension of WS-Trust1.3. As such, it does not have a **WSDL**.

WS-Trust 1.3 WSDL: The full WSDL for WS-Trust can be found at: <http://docs.oasis-open.org/ws-sx/ws-trust/200512/ws-trust-1.3.wsdl>.

WSTEP XML Schema: For the convenience of implementation, the **XML Schema** is provided here.

```
<xs:schema xmlns:xs="http://www.w3.org/2001/XMLSchema"
  xmlns:wstep="http://schemas.microsoft.com/windows/pki/2009/01/enrollment"
  targetNamespace="http://schemas.microsoft.com/windows/pki/2009/01/enrollment"
  elementFormDefault="qualified">

  <xs:import namespace="http://www.w3.org/XML/1998/namespace"
    schemaLocation="http://www.w3.org/2001/xml.xsd" />

  <xs:element name="DispositionMessage" type="wstep:DispositionMessageType" nillable="true" />
  <xs:complexType name="DispositionMessageType">
    <xs:simpleContent>
      <xs:extension base="xs:string">
        <xs:attribute ref="xml:lang" use="optional" />
      </xs:extension>
    </xs:simpleContent>
  </xs:complexType>
  <xs:element name="CertificateEnrollmentWSDetail" nillable="true"
    type="wstep:CertificateEnrollmentWSDetailType" />
  <xs:complexType name="CertificateEnrollmentWSDetailType">
    <xs:sequence>
      <xs:element minOccurs="0" maxOccurs="1" name="BinaryResponse" nillable="true"
        type="xs:string" />
      <xs:element minOccurs="0" maxOccurs="1" name="ErrorCode" nillable="true" type="xs:int" />
      <xs:element minOccurs="0" maxOccurs="1" name="InvalidRequest" nillable="true"
        type="xs:boolean" />
      <xs:element minOccurs="0" maxOccurs="1" name="RequestID" type="xs:string"
        nillable="true" />
    </xs:sequence>
  </xs:complexType>
  <xs:element name="RequestID" type="xs:string" nillable="true" />
</xs:schema>
```

7 Appendix B: Product Behavior

The information in this specification is applicable to the following Microsoft products or supplemental software. References to product versions include updates to those products.

The following table shows the relationships between Microsoft product versions or supplemental software and the roles they perform.

Windows Releases	Server Role	Client Role
Windows 7 operating system	No	Yes
Windows Server 2008 R2 operating system	Yes	Yes
Windows 8 operating system	No	Yes
Windows Server 2012 operating system	Yes	Yes
Windows 8.1 operating system	No	Yes
Windows Server 2012 R2 operating system	Yes	Yes
Windows 10 operating system	No	Yes
Windows Server 2016 operating system	Yes	Yes
Windows Server operating system	Yes	Yes
Windows Server 2019 operating system	Yes	Yes
Windows Server 2022 operating system	Yes	Yes
Windows 11 operating system	No	Yes
Windows Server 2025 operating system	Yes	Yes

Exceptions, if any, are noted in this section. If an update version, service pack or Knowledge Base (KB) number appears with a product name, the behavior changed in that update. The new behavior also applies to subsequent updates unless otherwise specified. If a product edition appears with the product version, behavior is different in that product edition.

Unless otherwise specified, any statement of optional behavior in this specification that is prescribed using the terms "SHOULD" or "SHOULD NOT" implies product behavior in accordance with the SHOULD or SHOULD NOT prescription. Unless otherwise specified, the term "MAY" implies that the product does not follow the prescription.

[<1> Section 3.1.3](#): Applicable Windows Server releases set the *DefaultLanguage* parameter to the installed language.

[<2> Section 3.1.4.1.3.2](#): Microsoft Windows always includes the requested end entity certificate in the **RequestedSecurityToken**.

8 Change Tracking

This section identifies changes that were made to this document since the last release. Changes are classified as Major, Minor, or None.

The revision class **Major** means that the technical content in the document was significantly revised. Major changes affect protocol interoperability or implementation. Examples of major changes are:

- A document revision that incorporates changes to interoperability requirements.
- A document revision that captures changes to protocol functionality.

The revision class **Minor** means that the meaning of the technical content was clarified. Minor changes do not affect protocol interoperability or implementation. Examples of minor changes are updates to clarify ambiguity at the sentence, paragraph, or table level.

The revision class **None** means that no new technical changes were introduced. Minor editorial and formatting changes may have been made, but the relevant technical content is identical to the last released version.

The changes made to this document are listed in the following table. For more information, please contact dochelp@microsoft.com.

Section	Description	Revision class
3.1.4.1.3.2 wst:RequestedSecurityTokenType	11199 : Updated to clarify the use of RequestSecurityTokenResponseCollection and RequestedSecurityToken response together with each containing the issued certificate; and clarified the return types for the BinarySecurityToken.	Major
Z Appendix B: Product Behavior	Added Windows Server 2025 to the list of applicable products.	Major

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